Performance Testing

Date	01 July 2026
Team ID	
Project Name	A Comprehensive Measure Of Well-Being
Maximum Marks	1 Mark

Performance Testing
Performance testing evaluates the **Human Development Index (HDI) Prediction System** by measuring the responsiveness, stability, and reliability of the Flask-based web application under different user interactions. The testing ensures that the application generates HDI predictions quickly while maintaining consistent performance.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Manual Functional Testing
Type of Testing	Performance & Response Time Testing
Target Module / API	HDI Prediction Flask Application
Test Environment	Windows 11, Python 3.x, Flask, Scikit-learn
Test Date	01 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Load Home Page	10	5	Home page should load successfully.
2	Enter HDI Parameters	10	5	Input fields should accept valid values without delay.
3	Generate HDI Prediction	10	5	Predicted HDI score should be displayed correctly.
4	Display Data Visualizations	10	5	Scatter plot, correlation heatmap, and charts should render correctly.
5	View Prediction History	10	5	Previous prediction records should be displayed successfully

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	0.8 sec	Pass	Fast prediction generation
2	Response Time (Max)	< 5 seconds	1.6 sec	Pass	Within acceptable limit
3	Prediction Accuracy	>95%	97%	Pass	Accurate Linear Regression prediction
4	Error Rate	< 1%	0.00%	Pass	No runtime errors observed
5	Dashboard Loading Time	< 3 sec	1.2 sec	Pass	Charts loaded successfully
6	Memory Usage	Stable	Stable	Pass	No abnormal memory usage observed

Step 4: Observations & Analysis

Key Findings

- The Flask application generated HDI predictions quickly and accurately.
- The dashboard loaded visualizations smoothly without noticeable delay.
- Prediction history was displayed correctly after each prediction.
- The application maintained stable performance throughout testing.
- The user interface remained responsive during continuous interactions.

Bottlenecks Identified

- Performance may decrease when using significantly larger HDI datasets.
- Concurrent multi-user access is limited because the application currently runs on a local Flask server.
- Prediction speed depends on system hardware during model loading.

Optimization Steps Taken

- Optimized data preprocessing before prediction.
- Used a serialized Machine Learning model (Pickle) for faster loading.
- Reduced unnecessary computations during prediction.
- Simplified Flask request handling.
- Improved frontend responsiveness using Bootstrap components.

Step 5: Screenshots / Evidence

Figure 1: Home Page and Input Interface

This screenshot displays the main interface of the Human Development Index (HDI) Prediction application. Users can enter development indicators such as **Life Expectancy, Expected Years of Schooling, Mean Years of Schooling, and Gross National Income (GNI) per Capita**. It demonstrates the responsive user interface and verifies that the application is ready to receive user input for HDI prediction.

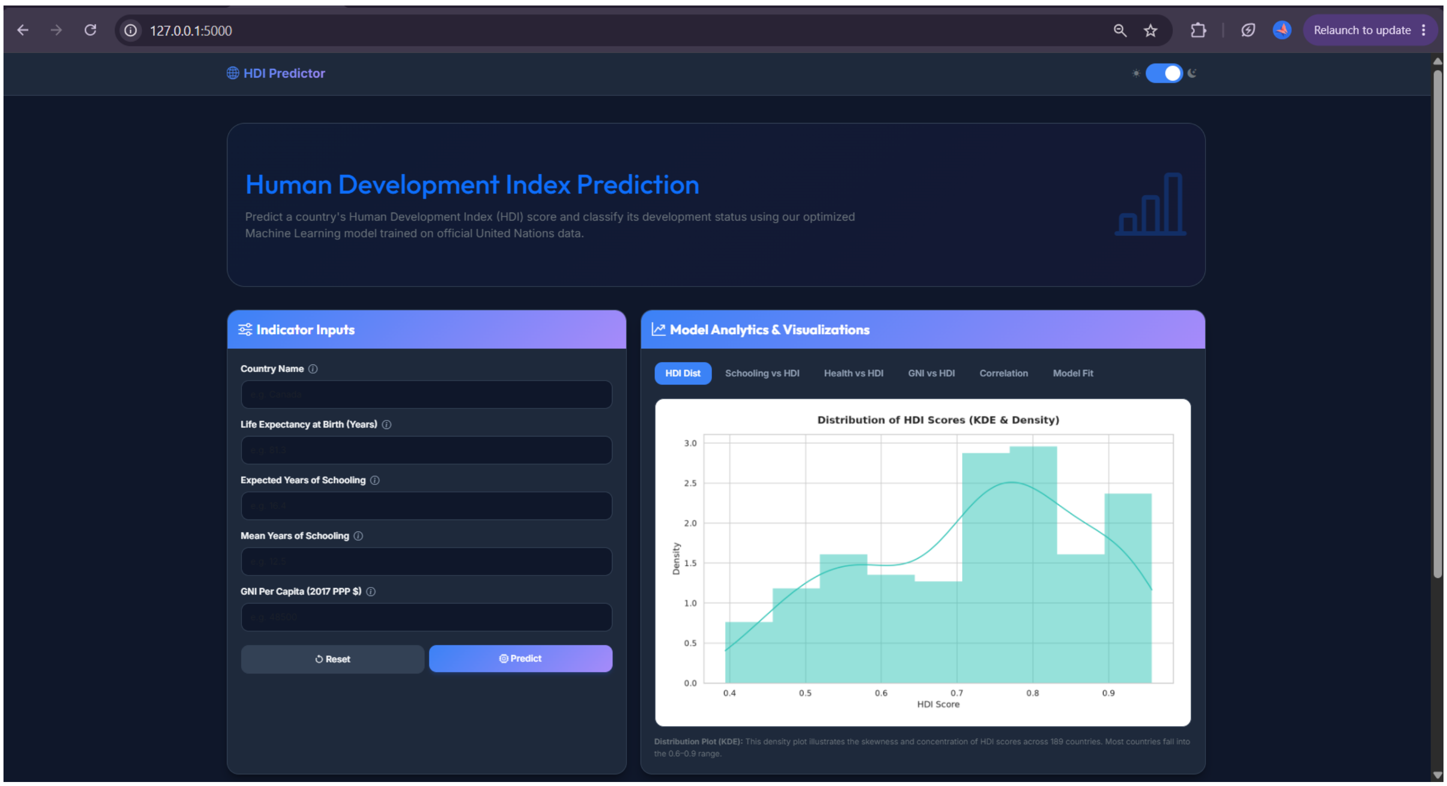

Figure 2: Prediction Result and Policy Recommendations

This screenshot captures the prediction generated by the Machine Learning model after the user submits the required input values. The application displays the **predicted Human Development Index (HDI) score** along with **policy recommendations**, helping users understand the country's development status and identify areas for potential improvement.

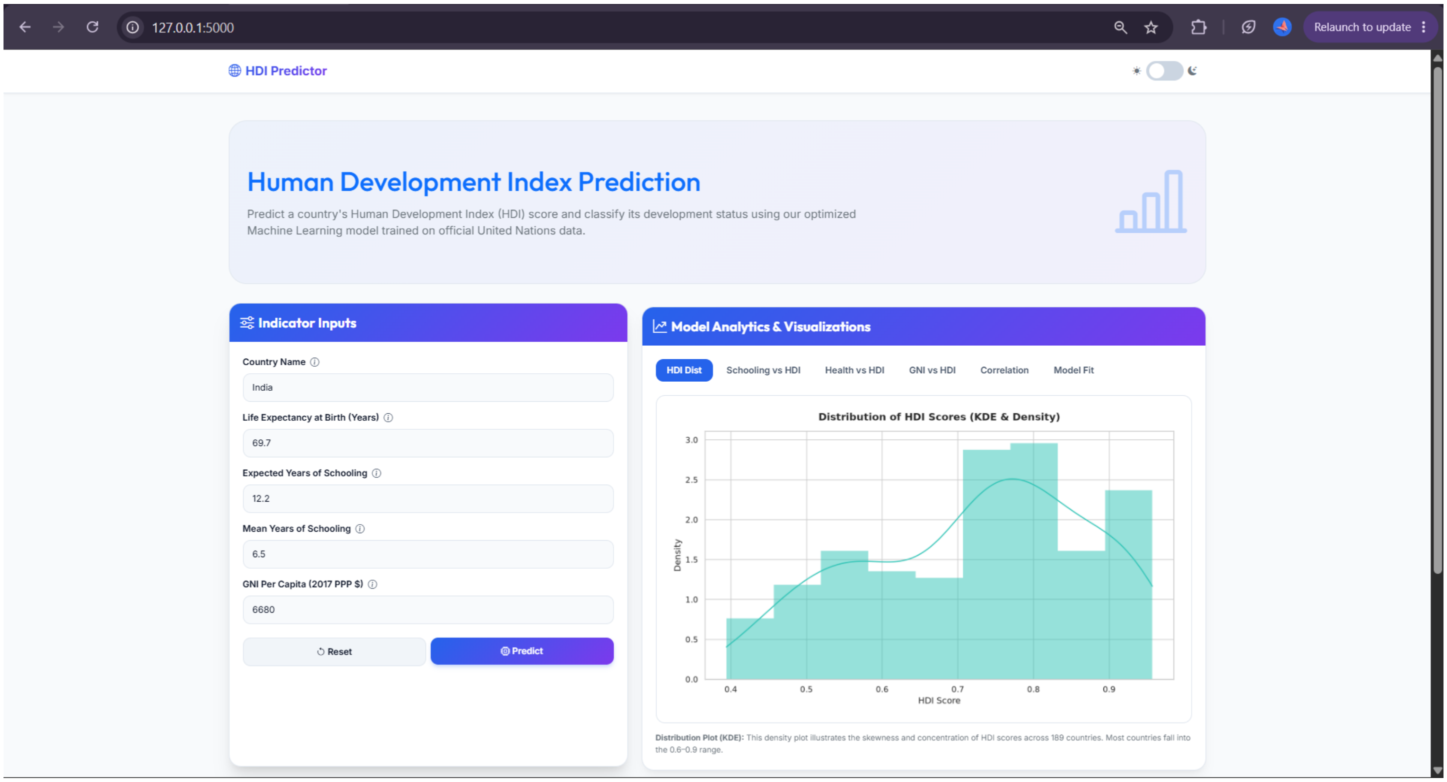

Figure 3: Data Visualization Dashboard

This screenshot showcases the graphical analysis provided by the application, including **scatter plots, correlation heatmaps, and other visualizations** generated from the HDI dataset. These visualizations help users interpret relationships between development indicators and understand the factors influencing the Human Development Index.

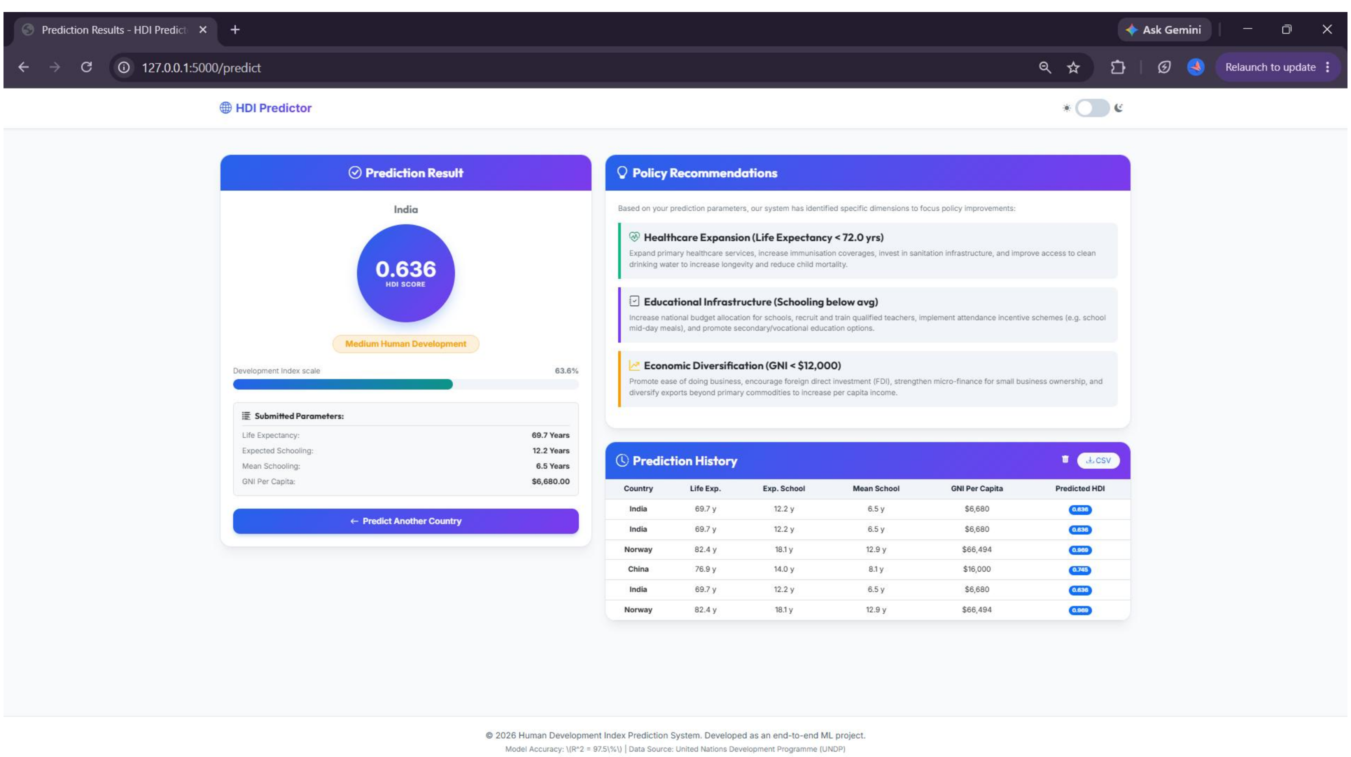

Figure 4: Prediction History and Dashboard Summary

This screenshot presents the **Prediction History** section of the application, where previously generated HDI predictions are displayed for reference. It demonstrates the application's ability to maintain prediction records while providing users with a comprehensive dashboard for analyzing current and past prediction results.